

A geometric derivation of the proton charge radius from a relational projection of a single event

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11 de abril

Abstract

We present a relational geometric framework based on a postulate of uniqueness and a rule of minimal binary differentiation. Under these assumptions, an internal theorem yields a discrete sequence of solutions $r_V = 2^{-\varepsilon}$ ($\varepsilon \in \mathbb{N}_{\geq 0}$) and an associated arithmetic structure $M_\varepsilon = 2^\varepsilon - 1$. The first compositional closure occurs at $\varepsilon = 4$, for which the projected scale is

$$R_V = \frac{4\hbar}{m_p c} \approx 0.841\,236 \text{ fm.}$$

We propose, as an *electromagnetic bridge hypothesis*, that this scale coincides with the proton root-mean-square charge radius. The equality $R_V = r_p$ is not a theorem of the framework: it is the physical hypothesis that connects the relational geometry with the electromagnetic observable, and it is not yet derived from a microscopic theory of the current. Under this hypothesis, the prediction agrees with CODATA 2018 ($r_p^{\text{exp}} = 0.8414(19)$ fm) at the 0.02% level and remains within 1σ of the CODATA/NIST 2022 recommended value ($r_p = 0.84075(64)$ fm). Under an additional maximum-entropy ansatz for the projected density, one obtains $G_E(Q^2) = \exp(-Q^2 R_V^2/6)$ with an independent prediction for the curvature $C_4 = R_V^4/72 \approx 6.96 \times 10^{-3} \text{ fm}^4 \approx 4.59 \text{ GeV}^{-4}$. The framework is presented as a falsifiable geometric correspondence, not as a replacement for QCD. Additional correspondences for the electron, free neutron, baryonic trivalence, spinorial behavior under double covering, and sub-baryonic mesons reinforce the internal coherence of the framework. We identify four remaining gaps: an absolute derivation of binary differentiation, a derivation of the bridge hypothesis from an effective electromagnetic current, an extension to heavy quarks, and a complete theory of mesons.

1 Epistemic status of the framework

Before presenting the formal development, we explicitly declare the status of each component. The table distinguishes four categories:

- **Postulate:** a foundational or selection assumption not derived within the framework.
- **Internal theorem:** a mathematical consequence of the postulates.
- **Bridge hypothesis:** an external physical identification, not derived from the framework, that connects an internal quantity with an observable.
- **Falsifiable prediction:** a consequence of a theorem or additional ansatz, testable against data.

Component	Status
Uniqueness of the event $\aleph$	Foundational postulate
Coherence of the present	Foundational postulate
Operationality of mathematics	Foundational postulate
Universal frequency $\omega_{\aleph}$ of the origin	Foundational postulate
Image frequency $\omega_{\mathcal{I}} = \omega_{\aleph} \cdot t_{\mathcal{I}}$	Definition
Pythagorean identity $\omega_{\mathcal{I}}^2 = \omega_V^2 + \omega_m^2$	Structural postulate
Minimal binary differentiation	Selection postulate
Quantization $r_V = 2^{-\varepsilon}$	Internal theorem (given the binary postulate)
Mersenne structure $M_{\varepsilon} = 2^{\varepsilon} - 1$	Internal algebraic theorem
First compositional closure ($\varepsilon^* = 4$)	Mathematical theorem
Assignment $\varepsilon = 4 \rightarrow$ proton	Selection criterion, not a complete physical derivation
Relation $R_V = 4\bar{\lambda}_p$	Internal theorem (if $\varepsilon = 4$ is accepted)
Identification $R_V = r_p^{\text{charge}}$	Electromagnetic bridge hypothesis
$G_E(Q^2) = e^{-Q^2 R_V^2/6}$	Effective low- Q^2 model under maximum entropy
Curvature prediction $C_4 = R_V^4/72$	Falsifiable prediction of the Gaussian ansatz
Baryonic trivalence	Modest internal result
Double covering $\Phi = e^{i\varphi/2}$	Conditional theorem ($SU(2) \rightarrow SO(3)$)
Exact relational non-duplication	Tautology under uniqueness; partial analogy with Pauli
Absolute derivation of binary differentiation	Open gap
Derivation of the bridge hypothesis	Open gap
Heavy quarks	Open gap
Complete theory of mesons	Open gap

2 Introduction

The proton charge radius is one of the most fundamental parameters of hadronic physics. Independent measurements via electron–proton scattering [3, 4] and muonic hydrogen spectroscopy [1, 2] converge, after the decade-long discrepancy known as the “proton radius puzzle,” on recommended values of $r_p = 0.8414(19)$ fm according to CODATA 2018 [5] and $r_p = 0.84075(64)$ fm according to CODATA/NIST 2022 [6]. Constituent quark models treat it as a fitting parameter, and lattice QCD calculations reproduce it with considerable computational effort but without revealing an underlying geometric structure [10].

In this work we propose a concrete geometric correspondence. The argument chain, organized explicitly in this document, is as follows: a foundational postulate of uniqueness (all images are

projections of the same origin $\aleph$), together with a postulate of minimal binary differentiation, generates an enumerable sequence of solutions whose arithmetic structure is dictated by the Mersenne numbers. Assigning the proton to the first compositional closure ($\varepsilon = 4$) allows the internal derivation of a scale $R_V = 4\hbar/(m_p c) \approx 0.8412$ fm. The identification of this scale with the proton's rms charge radius, $R_V = r_p^{\text{charge}}$, is not an internal theorem of the framework: it is an *electromagnetic bridge hypothesis* (Section 7) that connects a geometric quantity with a physical observable, and whose validity depends on additional tests — in particular, the curvature of the electromagnetic form factor, an independent prediction under a maximum-entropy ansatz for the projected density.

The framework presented here is not intended to replace quantum chromodynamics or the Standard Model. Its purpose is complementary: to offer a geometric description of the structural relations between types of elementary particles, where empirical regularities (number of quarks per baryon, boson–fermion distinction, non-duplication of identical fermions) emerge as consequences of the geometric architecture.

Equivalence with prior notation. Earlier versions of the framework used an index n such that $\omega_V = \omega_{\mathcal{I}}/2^n$ with $n = 2$ corresponding to the proton. The current notation employs ε with $\omega_V = \omega_{\mathcal{I}} \cdot 2^{-\varepsilon/2}$. The two notations are related by $\varepsilon = 2n$, with the proton case corresponding to $n = 2 \Leftrightarrow \varepsilon = 4$.

3 Foundational postulates

Postulate 1 (Uniqueness of the origin). *The origin proceeds from a single event $\aleph$, characterized by a universal angular frequency $\omega_{\aleph}$ and a projection relation c , both ontological limits as universal constants of the model. The totality of what is observable consists of images projected from $\aleph$ onto a relational space. The images are not separate entities: they are harmonics of $\aleph$, distinguished from one another by phase and harmonic depth of their projection.*

Postulate 2 (Coherence of the present). *All projected images share a single present, synchronized with the current phase of the origin $\aleph$. The existence of any image consists in its coherence with that shared present. Loss of coherence is the dissipation of the image.*

Postulate 3 (Operationality of mathematics). *The mathematical description of the model is operational: the equations define verifiable relations between measurable quantities. The question of whether such equations describe, represent, or constitute the underlying mechanism of coherence is considered open and is not resolved within the formal scope of the present work.*

4 Definitions

Definition 1 (Projected image). *A projected image of the origin $\aleph$ is a solution of the framework characterized by a pair*

$$\mathcal{I} = (\varepsilon, \varphi)$$

where $\varepsilon \in \mathbb{N}_{\geq 0}$ is the harmonic depth and $\varphi \in [0, 2\pi)$ is the relational phase of the image.

Definition 2 (Harmonic image frequency). *For each image $\mathcal{I}$, its characteristic harmonic frequency is the projection of the universal frequency $\omega_{\aleph}$ through the coupling factor $t_{\mathcal{I}}$ of the image with the shared present:*

$$\omega_{\mathcal{I}} \equiv \omega_{\aleph} \cdot t_{\mathcal{I}}. \tag{1}$$

For images with observable mass $m_{\mathcal{I}}$, the coupling factor relates to mass through the operational Compton relation:

$$\omega_{\mathcal{I}} = \frac{m_{\mathcal{I}}c^2}{\hbar}. \quad (2)$$

Remark 1 (Plurality of frequencies and uniqueness of the origin). $\omega_{\mathcal{I}}$ is not a frequency distinct from $\omega_{\mathbb{N}}$. It is the unique universal frequency of the origin, manifested in the specific relational context of the image $\mathcal{I}$ through the factor $t_{\mathcal{I}}$. The plurality of observable frequencies (ω_p , ω_e , ω_n , etc.) corresponds to a plurality of projections of the unique origin, not to a plurality of fundamental frequencies. The coupling factor $t_{\mathcal{I}}$ encodes how each image synchronizes with the cycle of the origin, and its numerical value for each image is fixed, in this work, by the observable mass via (2). An independent derivation of $t_{\mathcal{I}}$ from the structure of the framework constitutes future work.

Definition 3 (Modal decomposition). The harmonic frequency $\omega_{\mathcal{I}}$ of an image decomposes into two orthogonal components:

$$\omega_{\mathcal{I}}^2 = \omega_V^2 + \omega_m^2, \quad (3)$$

where ω_V is the projected mode and ω_m is the confined mode. The two components are not separate frequencies: they are the two orthogonal components of the unique frequency $\omega_{\mathcal{I}}$ that characterizes the image.

Remark 2 (Interpretation of the modes). By analogy, in a monochromatic electromagnetic wave with frequency ω , the electric and magnetic fields are orthogonal components of the same mode, not two separate modes. The frequency is one; the components are two. The component ω_V corresponds to aspects that project onto external references (transmitted in relations with other images and producing observables such as size or charge). The component ω_m corresponds to confined aspects (which sustain internal coherence and are not directly projected).

Remark 3 (Quadratic coherence criterion). Identity (3) implies

$$\omega_m^2 = \omega_{\mathcal{I}}^2 - \omega_V^2. \quad (4)$$

The sign of the discriminant $\Delta = \omega_{\mathcal{I}}^2 - \omega_V^2$ determines the nature of the confined mode:

- If $\Delta \geq 0$, ω_m is real and the image maintains oscillatory coherence with the shared present (stable existence as a free image).
- If $\Delta < 0$, ω_m is imaginary and coherence with the shared present becomes exponential damping (the image is not sustained as free existence, but as a transient state).

This quadratic criterion unifies, under a single mathematical principle, the stability of baryonic and leptonic images (Sections 6 and 8) with the instability of sub-baryonic mesons (Section 12).

Definition 4 (Integer weights). The modal decomposition (3) is parametrized by integer weights (W_V, W_m) via

$$r_V = \frac{W_V}{W_V + W_m}, \quad r_m = \frac{W_m}{W_V + W_m}, \quad (5)$$

with $r_V + r_m = 1$, and the identities

$$\omega_V = \omega_{\mathcal{I}}\sqrt{r_V}, \quad \omega_m = \omega_{\mathcal{I}}\sqrt{r_m}. \quad (6)$$

Definition 5 (Relational radius). To each modal component a characteristic radius is associated through the projection relation c :

$$R_V \equiv \frac{c}{\omega_V}, \quad R_m \equiv \frac{c}{\omega_m}. \quad (7)$$

5 Minimal binary differentiation and quantization

5.1 Differentiation postulate

Postulate 4 (Minimal binary differentiation). *Every elementary relational distinction between an image and the origin separates information into a complete system of two orthogonal channels: the projected channel and the confined channel. The completeness of the system is expressed by the quadratic conservation $\omega_V^2 + \omega_m^2 = \omega_I^2$, equivalent to $r_V + r_m = 1$. Each distinction halves the projected fraction.*

Comment. Minimal binary differentiation formalizes the idea that the individuation of an image proceeds through successive elementary choices, each with two alternatives that exhaust the decomposition of the frequency. The term *complete system* is used here in its standard technical sense from mathematical physics (analogous to a complete orthogonal basis or to a complete system of events in probability): the two channels cover the totality of the decomposition, with no residue in other channels. Any admissible minimal differentiation must be binary (an option and its complement) for the system to be complete without requiring additional information.

Remark 4 (Motivations for the binary postulate). *Postulate 4 is not derived from the previous postulates. However, two independent arguments motivate it structurally:*

(i) **Topological motivation.** *Minimal differentiation in a relational space does not admit three simultaneous alternatives as an elementary choice: a non-binary distinction requires additional parameters to be resolved. The most elementary form of differentiating an image from the origin is opposition (something and its complement), analogous to the fundamental dualities of physics (crest/trough of a wave, particle/antiparticle, positive/negative phase hemisphere). This motivation is ontological, not algebraic: it posits that binarity is a property of differentiation itself rather than a consequence of a more primary mathematical structure.*

(ii) **Representational motivation.** *For $\varepsilon = 4$ specifically, the fundamental representation of the algebra $su(2)$ governing spin-1/2 structure has complex dimension 2, equivalent to real dimension 4. The number $\varepsilon = 4$ is therefore the first harmonic depth compatible with the minimal storage of the non-commutative spin structure (reference, measurement axis, non-commuting generator, binary orientation). This motivation connects the binary postulate with the structure of the rotation group, reinforcing the assignment $\varepsilon = 4 \rightarrow p$ from a direction independent of the harmonic compositionality criterion of Theorem 3.*

Neither motivation constitutes a complete derivation. An absolute derivation of Postulate 4 from more fundamental structures remains as an open gap (Section 14, Gap 1).

Remark 5 (Structure vs. numerical determination). *Postulate 4 fixes the structure of the decomposition (complete system, orthogonal, with quadratic conservation) but does not by itself determine the specific values of r_V and r_m for a given image. Given a real value ω_I , the Pythagorean identity $\omega_V^2 + \omega_m^2 = \omega_I^2$ alone admits a continuum of solutions (ω_V, ω_m) . The unique determination of the components in terms of the harmonic depth ε is obtained from the recursive rule of Postulate 4 applied to an image with ε independent distinctions, formalized in Theorem 1. Uniqueness does not come from the Pythagorean identity alone, but from its composition with the recursive halving rule.*

5.2 Quantization as theorem

Theorem 1 (Binary quantization). *If an image requires ε independent relational distinctions to close its identity with respect to the origin, then the projected quadratic fraction is*

$$r_V = 2^{-\varepsilon}, \quad \omega_V = \omega_I \cdot 2^{-\varepsilon/2}. \quad (8)$$

Proof. Let $r_V(k)$ be the projected fraction after k distinctions, with $r_V(0) = 1$. By Postulate 4, $r_V(k+1) = r_V(k)/2$. By induction, $r_V(\varepsilon) = 2^{-\varepsilon}$. The identity for ω_V follows from Definition 4. $\square$

Modular lemma as arithmetic witness.

Lemma 1 (2–3 tuning via twin primes). *Let $(p, p+2)$ be a pair of twin primes with $p \geq 5$, and let $m = p+1$. Then $m \equiv 0 \pmod{6}$.*

Proof. For $p \geq 5$, neither p nor $p+2$ is divisible by 2 or 3. Therefore $p+1$ is divisible by 6. $\square$

Lemma 1 establishes an arithmetic affinity between the binary quantization and the structure of twin primes. It functions as a motivational witness, not as an additional derivation.

6 Derivation of the proton radius

Theorem 2 (Mersenne structure). *For every coherent image with depth $\varepsilon \in \mathbb{N}_{\geq 0}$,*

$$\frac{\omega_m^2}{\omega_V^2} = 2^\varepsilon - 1 \equiv M_\varepsilon. \quad (9)$$

Proof. From (3), $\omega_m^2 = \omega_I^2 - \omega_V^2$. From Theorem 1, $\omega_V^2 = \omega_I^2 \cdot 2^{-\varepsilon}$. Substituting, $\omega_m^2/\omega_V^2 = 2^\varepsilon - 1$. $\square$

6.1 Low-order spectrum of solutions

ε	M_ε	Factorization	New primes	Tentative correspondence
0	0	—	—	Null mode
1	1	unit	—	Symmetric case ($r_V = r_m$)
2	3	prime	{3}	Electron
3	7	prime	{7}	Free neutron
4	15	3 · 5	{5}	Proton
5	31	prime	{31}	Macroscopic transient state*
6	63	$3^2 \cdot 7$	—	Macroscopic collective closure*
7	127	prime (Mersenne)	{127}	Collapse threshold*

Table 1: Spectrum of solutions for $\varepsilon \leq 7$. Correspondences for $\varepsilon \in \{5, 6, 7\}$ marked with * are pointers to separate work in the research program (see Section 13); they are not developed in the present work. The assignment for $\varepsilon = 6$ corresponds to a many-body collective closure, not to an elementary particle: the signature $M_6 = 63 = 3^2 \cdot 7$ contains only inherited primes with no new prime contribution, an arithmetically exceptional configuration among $\varepsilon \leq 30$.

6.2 First compositional closure

Theorem 3 (First compositional closure). *Let ε^* be the smallest integer for which M_{ε^*} is a composite number whose factorization contains at least one prime previously present in the sequence $\{M_\varepsilon\}_{\varepsilon < \varepsilon^*}$ and at least one new prime. Then $\varepsilon^* = 4$.*

Proof. Direct verification of Table 1: $\varepsilon \leq 3$ are trivial or prime without combination; $\varepsilon = 4$ with $M_4 = 3 \cdot 5$ has inherited prime $\{3\}$ and new prime $\{5\}$. $\square$

Correspondence 1 (Proton assignment). *The image $\varepsilon = 4$ is assigned to the proton. This assignment is not asserted as a complete derivation of the physical identity of the proton: it is proposed as the minimal assignment compatible with (a) the first stable composite structure that the framework selects (Theorem 3), and (b) the quantitative coincidence with the observed electromagnetic scale of the proton under the bridge hypothesis (Section 7). Its validity depends on additional independent predictions, in particular the curvature of the form factor (Section 11).*

6.3 Relation for the proton radius

Theorem 4 (Relational radius of the proton). *For the image $\varepsilon = 4$ assigned to the proton (Correspondence 1),*

$$R_V = 4 \bar{\lambda}_p = \frac{4\hbar}{m_p c} \approx 0.841\,236 \text{ fm}. \quad (10)$$

Proof. By Theorem 1 with $\varepsilon = 4$: $\omega_V = \omega_p/4$, where $\omega_p \equiv \omega_{\mathcal{I}}^{(p)} = m_p c^2/\hbar$. By Definition 5: $R_V = c/\omega_V = 4\hbar/(m_p c) = 4\bar{\lambda}_p$. $\square$

6.4 Empirical verification

Using $\bar{\lambda}_p = 2.103\,089\,103 \times 10^{-16} \text{ m}$ (CODATA 2018), $R_V = 0.841\,236 \text{ fm}$.

Comparison with CODATA 2018. With $r_p^{\text{exp}} = 0.8414(19) \text{ fm}$ [5]:

$$\frac{R_V - r_p^{\text{exp}}}{r_p^{\text{exp}}} \approx -1.9 \times 10^{-4} \approx -0.02\%. \quad (11)$$

Comparison with CODATA/NIST 2022. With $r_p^{\text{exp}} = 0.84075(64) \text{ fm}$ [6], the absolute deviation is $|R_V - r_p^{\text{exp}}| \approx 0.000486 \text{ fm}$, or equivalently $\sim 0.76\sigma$. The prediction remains within 1σ of the most recent recommended value.

6.5 Dimensionless constant κ

Defining $\kappa \equiv m_p c r_p / \hbar$, the framework predicts $\kappa = 4$ exactly. Experimentally:

- With CODATA 2018: $\kappa^{\text{exp}} \approx 4.0008$.
- With CODATA/NIST 2022: $\kappa^{\text{exp}} \approx 3.9977$.

Both values are compatible with $\kappa = 4$ within experimental uncertainty.

7 Electromagnetic bridge hypothesis

Theorem 4 establishes that, under binary quantization and the assignment $\varepsilon = 4 \rightarrow p$, the projected scale of the framework is $R_V = 4\bar{\lambda}_p$. This is an internal geometric quantity of the framework, mathematically derived from the postulates. But the physical observable of interest — the proton's rms charge radius r_p^{charge} extracted from electromagnetic experiments — is not an internal quantity

of the framework: it belongs to the proton's electromagnetic current, a physical structure external to the geometric framework presented here.

The connection between R_V and r_p^{charge} is not derived, in this work, from the previous postulates. It is introduced explicitly as an additional hypothesis:

Postulate 5 (Electromagnetic bridge hypothesis). *Let J^μ be the effective electromagnetic current associated with the proton image. In the non-relativistic low-momentum-transfer limit, the projected component V determines the second moment of the electric charge density:*

$$\langle r^2 \rangle_E \equiv -6 \left. \frac{dG_E}{dQ^2} \right|_{Q^2=0} = R_V^2. \quad (12)$$

Equivalently, $r_p^{\text{charge}} = R_V$.

Epistemic status of the bridge hypothesis. The equality $R_V = r_p^{\text{charge}}$ is not a theorem of the framework: it is the physical hypothesis that connects the relational geometry with the electromagnetic observable. Under this hypothesis, the numerical verifications of Subsection 6.4 hold. Without this hypothesis, the framework produces a scale R_V close to the measured charge radius but does not establish their identity.

Why V would be electromagnetic: open question. Postulate 5 declares that the projected mode V is the one that couples to the photon as electric charge. *Why this should be the case is not derived from the present framework.* A complete derivation would require constructing the effective electromagnetic current from the structure of the framework, showing that V naturally couples to the electromagnetic field while m does not. This derivation is not carried out in the present work and constitutes one of the remaining gaps (Section 14).

Falsifiability of the bridge hypothesis. The bridge hypothesis as stated in (12) fixes only the second moment of $G_E(Q^2)$. By itself, it predicts the slope of G_E at the origin but not its global form. The independent falsifiable prediction of the framework comes from the combination of the bridge hypothesis with an additional ansatz on the functional form of the projected relational density (Section 11). The non-trivial quantitative test is the curvature C_4 of the form factor at low Q^2 .

8 Additional correspondences

8.1 Electron ($\varepsilon = 2$)

Theorem 5 (Electron radii). *For $\varepsilon = 2$ assigned to the electron,*

$$R_V^{(e)} = 2 \bar{\lambda}_e, \quad R_m^{(e)} = \frac{2}{\sqrt{3}} \bar{\lambda}_e, \quad (13)$$

with $\bar{\lambda}_e \approx 386.16 \text{ fm}$.

Physical interpretation. The radius $R_V^{(e)}$ is not a charge radius nor a classical spatial extent, and the operational correspondence $R_V \leftrightarrow r_p^{\text{charge}}$ for the proton *does not* automatically extend to the electron. For the electron, $R_V^{(e)}$ is a coherence/projection scale associated with the Compton

wavelength, where the notion of a classical point particle ceases to apply and pair creation becomes kinematically accessible. This asymmetry (charge radius for proton, Compton scale for electron) reflects structural differences between the two images (composite vs. prime signature) that require additional rules not derived in the present work.

8.2 Free neutron ($\varepsilon = 3$)

Theorem 6 (Free neutron radii). *For $\varepsilon = 3$,*

$$R_V^{(n)} = 2\sqrt{2} \bar{\lambda}_n, \quad R_m^{(n)} = \frac{2\sqrt{2}}{\sqrt{7}} \bar{\lambda}_n. \quad (14)$$

The harmonic signature $M_3 = 7$ is prime.

Differential harmonic stability. The proton's signature ($M_4 = 15 = 3 \cdot 5$) admits rational decomposition, allowing internal redistribution between sub-modes. The free neutron's signature ($M_3 = 7$, prime) does not. This structural correspondence is consistent with the observed instability of the free neutron versus the stability of the proton, but does not constitute a physical explanation of the instability: the beta decay of the neutron is determined by the neutron-proton-electron-antineutrino mass difference and the weak interaction. The harmonic rigidity of the framework is offered as a structural analogy, not as a causal mechanism.

9 Baryonic trivalence

Definition 6 (Subharmonics). *For an image with composite M_ε , the subharmonics of the confined mode are*

$$\omega_m^{(d)} \equiv \frac{\omega_m}{d}, \quad d \mid M_\varepsilon. \quad (15)$$

Theorem 7 (Trivalence of the proton). *For $\varepsilon = 4$, the first non-trivial subharmonic $\omega_m/3$ defines a substructure of three correlated sub-modes phase-shifted by intervals of $2\pi/3$. The associated equivalent mass is*

$$m_q = \frac{\sqrt{15}}{12} m_p \approx 302.83 \text{ MeV}/c^2. \quad (16)$$

Trivalence argument. The subharmonic ω_m/d requires d phase-shifted copies in intervals $2\pi/d$ to sustain coherence with the shared present. For $d = 3$ (first non-trivial divisor of M_4), this forces a trivalent substructure.

Modest quantitative correspondence. The value $m_q \approx 302.83 \text{ MeV}/c^2$ falls within the phenomenological scale of light constituent mass in the quark model ($\sim 313 \text{ MeV}/c^2$, error $\sim 3\%$). Constituent masses are phenomenological parameters with uncertainty of order 5–10% depending on source.

10 Spinorial behavior

10.1 Hemispheres and orientation

Definition 7 (Hemispheres). *For an image with $\varepsilon \geq 2$ (asymmetric signature), the phase φ decomposes into*

$$H_+ = \{\varphi : \sin \varphi > 0\}, \quad H_- = \{\varphi : \sin \varphi < 0\}.$$

For $\varepsilon \in \{0, 1\}$ the hemispheres are geometrically equivalent.

Definition 8 (Discrete orientation). For $\varepsilon \geq 2$, $\sigma \equiv \text{sgn}(\sin \varphi) \in \{+1, -1\}$.

10.2 Double covering as conditional theorem

Theorem 8 (Relational double covering). *Let an image with $\varepsilon \geq 2$ preserve an internal orientation $\sigma \in \{+1, -1\}$ that is not directly observable, whose inversion changes the relational state. The observable space of orientations is $SO(3)$. Then the minimal space covering the internal orientations is the double cover $SU(2)$, and the relational phase transforms as*

$$\Phi = e^{i\varphi/2}, \quad (17)$$

with $\Phi \rightarrow e^{i\theta/2}\Phi$ under frame rotation by θ .

Proof. The internal orientation σ must be represented in a state space that distinguishes $+1$ from -1 and projects onto $SO(3)$. The universal double cover of $SO(3)$ is $SU(2)$, with homomorphism $\pi : SU(2) \rightarrow SO(3)$ of kernel $\{+I, -I\}$. A frame rotation by 2π in $SO(3)$ corresponds to multiplication by $-I$ in $SU(2)$, which inverts σ . The relational phase must belong to the subgroup $U(1) \subset SU(2)$ generated by the angular coordinate of the internal state; its action on the state under frame rotation θ is $e^{i\theta/2}$. $\square$

Scope of Theorem 8. The theorem derives the relational double covering from the conditional hypothesis that an internal orientation σ exists, not directly observable, whose inversion affects the relational state. *Complete equivalence with physical spin 1/2 in relativistic Hilbert spaces is not obtained here:* physical fermionic behavior requires the spin-statistics theorem, representations of the Poincaré group, and quantum field structure. The present theorem establishes the minimal geometric condition (double covering) that such constructions require, not the constructions themselves.

10.3 Boson–fermion distinction

Theorem 9 (Relational boson–fermion category). *Images with $\varepsilon \in \{0, 1\}$ do not admit non-trivial double covering: the relational phase can be taken as $\Phi = e^{i\varphi}$ (angular coordinate without internal structure σ), and under $\theta = 2\pi$, Φ returns to its original value. They correspond to the bosonic relational category. Images with $\varepsilon \geq 2$ admit double covering due to the presence of σ (Theorem 8) and correspond to the fermionic relational category.*

On the nature of Φ . In the bosonic case, Φ is an observable relational angular coordinate. In the fermionic case, Φ lives in a double cover space due to the presence of the internal state σ . This distinction (internal structure present vs. absent) is what produces the difference in behavior under rotation.

10.4 Exact relational non-duplication

Theorem 10 (Exact relational non-duplication). *Two images with the same depth $\varepsilon \geq 2$ in mutual relation cannot simultaneously share: the same φ , the same σ , and the same local relational context.*

Proof. By Postulate 1, all images of the same type are the same unique solution of the framework. If two images coincided in all relational dimensions, they would be the same projection, not two distinguishable projections. $\square$

Partial correspondence with the exclusion principle. Theorem 10 establishes a geometric property of the framework: two projections of the same type cannot be identical in all relational identifiers without being one. This property is a *necessary condition* for the Pauli exclusion principle, but it is not *equivalent*: the complete Pauli principle is stated over quantum states in antisymmetric Hilbert spaces for identical fermions, which requires additional structure (second quantization, spin-statistics theorem). The present framework provides the geometric substrate of non-duplication; the complete quantum formulation remains as an unrealized extension.

11 Operational prediction on the form factor

11.1 Relational density from maximum entropy

The correspondence $R_V \leftrightarrow r_p^{\text{charge}}$ requires a relational density $\rho_V(\vec{r})$ that reproduces, via Fourier transform, the measured form factor. The functional form of ρ_V is not fixed by the previous postulates. We posit that the relational density must be the *minimally informative* compatible with the constraints of the framework: the maximum entropy principle [12].

Theorem 11 (Minimal relational density). *Let $\rho_V(\vec{r})$ be the relational density associated with the projected mode, subject only to:*

- *Normalization:* $\int \rho_V(\vec{r}) d^3r = 1$.
- *Second moment:* $\int r^2 \rho_V(\vec{r}) d^3r = R_V^2$.
- *Non-negativity:* $\rho_V(\vec{r}) \geq 0$.

The maximum entropy distribution under these constraints is isotropic Gaussian:

$$\rho_V(\vec{r}) = \left(\frac{3}{2\pi R_V^2} \right)^{3/2} \exp \left(-\frac{3r^2}{2R_V^2} \right). \quad (18)$$

Proof. Functional maximization of $-\int \rho_V \ln \rho_V d^3r$ with Lagrange multipliers for the normalization and second moment constraints yields the standard Gaussian form [12]. Isotropy follows from the fact that the only non-trivial constraint is radial. $\square$

Interpretation and scope. The Gaussian form is the unique density compatible with the minimal information that the framework provides about ρ_V . Two important notes:

1. The second moment $\langle r^2 \rangle = R_V^2$ is imposed as a constraint, not derived. Therefore, Theorem 11 does not independently prove that the physical second moment of the proton's charge distribution is R_V^2 ; it produces the *minimal functional form* compatible with that hypothesis.
2. The interpretation of $G_E(Q^2)$ as the Fourier transform of a spatial density should be understood as an effective approximation in the low- Q^2 limit, or in the kinematic frame appropriate for radius extraction. It does not constitute an exact relativistic spatial density of the proton, where the density-form factor relation involves recoil corrections and frame dependence [7, 8].

11.2 Form factor and curvature prediction

Corollary 1 (Gaussian form factor). *Under Theorem 11, in the low- Q^2 effective frame,*

$$G_E(Q^2) = \exp \left(-\frac{Q^2 R_V^2}{6} \right). \quad (19)$$

Low- Q^2 expansion.

$$G_E(Q^2) = 1 - \frac{Q^2 R_V^2}{6} + \frac{Q^4 R_V^4}{72} + \mathcal{O}(Q^6). \quad (20)$$

Comparing with the standard expansion $G_E(Q^2) = 1 - Q^2 \langle r^2 \rangle / 6 + C_4 Q^4 + \dots$:

- The first term reproduces $\langle r^2 \rangle_E = R_V^2$ (by construction).
- The second term *predicts*

$$C_4 = \frac{R_V^4}{72} \approx 6.96 \times 10^{-3} \text{ fm}^4 \quad (21)$$

for $R_V = 4\bar{\lambda}_p$.

Unit convention. Expression (21) uses natural units with Q expressed in fm^{-1} , so that C_4 has units of fm^4 . If experimental data are expressed with Q^2 in GeV^2 , the conversion $\hbar c = 0.1973 \text{ GeV}\cdot\text{fm}$ must be applied:

$$C_4^{[\text{GeV}^{-4}]} = C_4^{[\text{fm}^4]} \cdot (\hbar c)^{-4} \approx 6.96 \times 10^{-3} \cdot (0.1973)^{-4} \text{ GeV}^{-4} \approx 4.59 \text{ GeV}^{-4}. \quad (22)$$

Status of the prediction. Prediction (21) is falsifiable against e - p scattering data at low Q^2 [3, 7]. If the data require a curvature substantially different from $R_V^4/72$, the maximum entropy principle applied to ρ_V is refuted, without implying that $R_V = 4\bar{\lambda}_p$ is incorrect. A deviation would indicate that the current postulates of the framework do not specify enough constraints on ρ_V and that there is additional structure of the proton image not captured by the second moment.

12 Partial mesonic criterion

Mesons, with modal decompositions of type $W_V : W_m \neq 1 : (2^\varepsilon - 1)$, do not correspond to solutions of Theorem 1. This section formulates a programmatic criterion, not a complete theory.

Definition 9 (Calibrated mesonic state). *A mesonic state with mass m_M has its projected component identified with $\omega_V^{(p)} = \omega_p/4 \approx 3.564 \times 10^{23} \text{ rad/s}$, and its confined component is determined from $\omega_{\mathcal{I}}^{(M)2} = \omega_V^{(p)2} + \omega_m^{(M)2}$, where $\omega_{\mathcal{I}}^{(M)} = m_M c^2 / \hbar$.*

Theorem 12 (Conditional mesonic criterion). *Under Definition 9, a state with $\omega_{\mathcal{I}}^{(M)} < \omega_V^{(p)}$ produces $\omega_m^{(M)} \in i\mathbb{R}$.*

Physical interpretation. An imaginary confined frequency, $\omega_m \in i\mathbb{R}$, mathematically corresponds to exponential time behavior $\propto e^{|\omega_m|t}$ or $\propto e^{-|\omega_m|t}$ instead of oscillatory. In the framework of Postulate 2, this is interpreted as loss of coherence with the shared present (damping, decay) instead of stable oscillation, anchoring with the observed instability of pions as virtual mediators of the residual nuclear force.

Connection with projective duality. This reading is consistent with the projective duality hypothesis explored in earlier work of the program, where an imaginary confined frequency is interpreted as existence in a complementary geometric plane (non-projected), not as non-existence. In this reading, pions do not cease to exist when $\omega_m \in i\mathbb{R}$: they exist in a distinct relational modality, complementary to the baryonic modality, which is consistent with their phenomenological role as mediators of nucleon interactions. The formal development of this hypothesis is not completed in the present work.

Meson	m_M (MeV/ c^2)	$\omega_I^{(M)}$ (rad/s)	$\omega_m^{(M)}$	Category
$\pi^\pm$	139.57	2.12×10^{23}	imaginary	sub-baryonic
π^0	134.98	2.05×10^{23}	imaginary	sub-baryonic
$K^\pm$	493.68	7.50×10^{23}	real	supra-baryonic
η	547.86	8.32×10^{23}	real	supra-baryonic
ρ	775.26	1.18×10^{24}	real	supra-baryonic

Table 2: Application of the calibrated mesonic criterion.

Explicit limitations and future direction. The choice of $\omega_V^{(p)}$ as reference is calibrated, not derived. Mesonic masses are used as inputs. Lifetimes are not predicted. A concrete future direction is to reformulate mesons as *relational dipoles* between an image and an anti-image, with a cancellation rule of the form $r_m(M_{q\bar{q}}) = r_{m,q} + r_{m,\bar{q}} - 2\chi_{q\bar{q}}$, where $\chi_{q\bar{q}}$ encodes the relational cancellation between opposite orientations. This direction is not developed in the present work and constitutes separate future work.

13 Hierarchy of harmonic closures: pointer to separate work

The Mersenne arithmetic structure $M_\varepsilon = 2^\varepsilon - 1$ of Theorem 2 continues generating solutions beyond $\varepsilon = 4$. An arithmetic classification of the first 30 values reveals three qualitatively distinct regimes according to the prime contribution to the signature:

- **Type A — Pure new prime** (M_ε prime, no inherited primes). Contributes a new orthogonal direction to the accumulated relational space, without combination. Cases: $\varepsilon \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, \dots\}$.
- **Type B — Mixed composite** (M_ε composite with at least one new prime and at least one inherited). Contributes a new direction and combines with previous directions. Cases: $\varepsilon \in \{4, 8, 9, 10, 12, 14, 15, 16, 18, 20, \dots\}$.
- **Type C — Pure repetition** (M_ε composite with no new prime, all primes inherited). Contributes no new orthogonal direction to the accumulated relational space. Unique case among $\varepsilon \leq 30$: $\varepsilon = 6$, with $M_6 = 63 = 3^2 \cdot 7$ (primes $\{3, 7\}$, both inherited from $\varepsilon = 2, 3$).

Positions $\varepsilon \in \{5, 6, 7\}$: pointer to separate work. The tentative correspondences in Table 1 for $\varepsilon = 5, 6, 7$ are developed in separate work of the same research program. There, the macroscopic correspondences associated with each arithmetic regime are explored (transient configurations in stellar collapse, many-body collective closures in compact matter, gravitational collapse transition thresholds). The present paper does not develop these correspondences and does not assume their results; it limits itself to recording that the hierarchy $\varepsilon = 4, 5, 6, 7$ admits such tentative identifications in the broader program.

The arithmetic singularity of $\varepsilon = 6$. It is worth noting, within the scope of the present paper, that $\varepsilon = 6$ is mathematically exceptional: it is the only position among $\varepsilon \leq 30$ with a pure repetition signature. This arithmetic singularity suggests that $\varepsilon = 6$ does not correspond to an additional elementary image in the sequence $\{2, 3, 4\} \rightarrow \{\text{electron, free neutron, proton}\}$, but to a qualitatively different closure. Its concrete physical correspondence falls outside the scope of this work and is addressed in the separate document mentioned.

14 Remaining gaps

Gap 1: absolute derivation of binary differentiation. Postulate 4 reformulates the quantization $r_V = 2^{-\varepsilon}$ as a theorem from a more primary postulate, but that more primary postulate remains a postulate, motivated by two independent arguments (Remark 4) but not derived. The derivation of binary differentiation from an even more fundamental structure remains pending. At least two exploratory routes have been considered: (a) self-mirror image of the origin's cycle, where only projections that maintain closed phase coherence under reflection in the origin would be binary divisions; (b) stability under iterated double projection, where the closure of the subspace generated by modes (U, V, m, a) under successive applications of the projection operator would impose dyadic restrictions on R . Route (b) has been technically analyzed and does not close with the current algebraic postulates: the subspace $\langle \hat{N}, U, V, a \rangle$ turns out to be invariant under double projection for all R , without restricting the discretization. The absolute derivation still requires an additional postulate or a more fundamental structure than what is currently available. Additionally, the coupling factor $t_{\mathcal{I}}$ of Definition 2 is fixed by the observable mass of each image via (2); an independent derivation of $t_{\mathcal{I}}$ from the structure of the framework constitutes future work.

Gap 2: derivation of the electromagnetic bridge hypothesis. Postulate 5 declares, without deriving it from the framework, that the projected mode V determines the second moment of the proton's charge density. The underlying question — *why V couples to the photon as electric charge while m does not* — remains open. A complete resolution would require constructing, within the framework, an effective electromagnetic current J_{eff}^μ associated with the proton image, demonstrating that its form factor satisfies $\langle r^2 \rangle_E = R_V^2$, and identifying mode V as the carrier of the electric charge property. This gap is probably the most important of the program: while it is not closed, $R_V = r_p^{\text{charge}}$ remains an external hypothesis, not a consequence of the geometric framework.

Gap 3: heavy quarks. Theorem 7 produces the constituent mass of the u/d quark via subharmonic of the proton, but it does not extend to higher generations (s, c, b, t). The subsequent divisors of $M_4 = 15$ do not correspond to known quarks. A possible extension via quadratic composition of effective weights $r_{m,i}$ (calibration with p, n, Ω^- and prediction of additional baryons) constitutes a separate work program.

Gap 4: complete theory of mesons. Theorem 12 offers a conditional criterion but not a complete theory. The direction of mesons as relational dipoles (Section 12) is a suggested route but not developed.

Closure strategy. The present work closes two of the five original gaps of the program: the relational density (via maximum entropy, Theorem 11) and the rotational phase (via conditional double covering, Theorem 8). Binary quantization has been reduced to a more primary postulate (minimal binary differentiation), but its absolute derivation remains as a foundational gap (Gap 1). A new gap, previously implicit, has been made explicit and named: the derivation of the electromagnetic bridge hypothesis from an effective current of the framework (Gap 2). Extensions to heavy quarks (Gap 3) and mesons (Gap 4) are conceived as separate programs: the first via quadratic composition of effective weights $r_{m,i}$ with calibration-prediction protocol; the second via dipolar formulation with relational cancellation. The gaps do not constitute a list of shortcomings of the current framework but a research plan with falsification protocols already identified.

15 Conclusions

We have presented a geometric framework based on a postulate of ontological uniqueness of an event $\aleph$ with universal frequency $\omega_{\aleph}$, of which the totality of what is observable consists of harmonic projections. Each image $\mathcal{I}$ has a characteristic frequency $\omega_{\mathcal{I}} = \omega_{\aleph} \cdot t_{\mathcal{I}}$ that is the unique universal frequency manifested in the specific relational context of the image. On four postulates internal to the framework (uniqueness, coherence, operationality, minimal binary differentiation), theorems are derived that articulate: the quantization $r_V = 2^{-\varepsilon}$; the Mersenne arithmetic structure; the first compositional closure at $\varepsilon = 4$; the proton assignment $\varepsilon = 4 \rightarrow p$ as a minimal selection criterion; the relation $R_V = 4\bar{\lambda}_p$ as an internal theorem; additional correspondences for the electron and free neutron; baryonic trivalence with u/d constituent mass at $\sim 3\%$; relational double covering as a conditional theorem from $SU(2) \rightarrow SO(3)$; and exact relational non-duplication as a necessary geometric condition for the exclusion principle.

The structure of the central claim. The central numerical result of the work — $R_V = 4\hbar/(m_p c) \approx 0.8412$ fm, compatible with the proton’s rms charge radius within experimental uncertainty — is obtained as the composition of two epistemically distinct steps:

1. *Internal theorem:* given the geometric framework and the assignment $\varepsilon = 4 \rightarrow p$, $R_V = 4\bar{\lambda}_p$.
2. *Bridge hypothesis:* $R_V = r_p^{\text{charge}}$ (Postulate 5).

The equality $R_V = r_p^{\text{charge}}$ is not a theorem of the framework: it is the physical hypothesis that connects the relational geometry with the electromagnetic observable. The dimensionless constant $\kappa = m_p c r_p / \hbar = 4$ that emerges from this composition, against the experimental value $\kappa^{\text{exp}} \in \{4.0008, 3.9977\}$ according to CODATA 2018 or 2022 respectively, constitutes a non-trivial quantitative correspondence. The prediction $C_4 = R_V^4/72 \approx 6.96 \times 10^{-3} \text{ fm}^4 \approx 4.59 \text{ GeV}^{-4}$ exposes the model to an independent quantitative test, falsifiable against e - p scattering data at low Q^2 .

Remaining gaps. The framework presents four declared gaps: (1) absolute derivation of binary differentiation; (2) derivation of the electromagnetic bridge hypothesis from an effective current of the framework; (3) extension to heavy quarks; (4) complete theory of mesons. The second is probably the most important: making it explicit as a named gap (instead of keeping it as an implicit operational correspondence) is one of the methodological contributions of the present work with respect to earlier versions of the program.

The framework does not intend to replace the Standard Model or quantum chromodynamics. Its contribution is complementary: it offers a geometric architecture where known empirical regularities (binary quantization of particle types, baryonic trivalence, boson-fermion distinction, non-duplication of fermions) emerge as derived consequences, suggesting an underlying structure of harmonic relations that standard theories do not explicitly expose. The non-trivial quantitative correspondence shown for the case of the proton radius, conditional on the bridge hypothesis, indicates that this structure deserves further investigation.

Acknowledgements

The editorial assistance received during the preparation of this manuscript is gratefully acknowledged.

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